pingexpress
Ping to send, Ping to receive

07/01/2017. Transportation Model.

A model is an abstraction of reality or reality to be.

Transportation is a means that describe various means or method of transporting goods from different sources or origin to its destination. Sources or origin is the production or storage point while destination is the point of demand.

The objective of transportation model is to minimize cost. For a transportation model to be solved, it must be balanced.

A balanced transportation model is a transportation model that is made up of equal demand & supply i.e when what is to be supplied at various origin is equal to what is been demanded at various destination otherwise, it is unbalanced.

Note: In reality, there's hardly a balanced transportation model.

To resolve an unbalanced transportation model, we create a dumie row or column to absorb the difference between demand & supply.

A dumie is a row or column (in transportation model) created with a zero cost of transportation to absorb the difference between demand & supply in order to make an unbalanced model balanced.

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

* Methods of Solving Transportation Models.

There are two methods:

(1) Method basic feasible solution method which Comprise of 3 means which are:

(a) North - west Corner method (b) Vogel approximation method (VAM)

(b) Least Cost method (LCM)

(c) Optimal method: This solution is made up of 2 means viz

* Stepping stone algorithm method

(b) Modified distribution method.

Illustration - A Company has 3 production facilities X_1, X_2 & X_3 with production Capacity of 700, 900 & 500 tons of their Product per week respectively. These units are to be shipped to 4 warehouses D_1, D_2, D_3 & D_4 with requirements of 500, 800, 700 & 400 units/week respectively. The transportation Cost in ₦ per units between factories and warehouses are given in the table below.

	D_1	D_2	D_3	D_4	Supply (tons)
X_1	19	30	50	10	700
X_2	70	30	40	60	900
X_3	40	8	70	20	500
Demand	500	800	700	400	2400

(a) Formulate as linear programming model

(b) Solve for initial basic feasible solution

(c) Calc. the cost of

Solution & Compare

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

Note: There are 4 steps in formulating a linear programming model

(i) Identify the decision variable

(ii) Formulate or state the objective function

(iii) Apply the constraints

(iv) State the non-negativity function.

Soln

	D_1	D_2	D_3	D_4	
X_1	S_{11}	S_{12}	S_{13}	S_{14}	700
X_2	S_{21}	S_{22}	S_{23}	S_{24}	900
X_3	S_{31}	S_{32}	S_{33}	S_{34}	500
dd	500	800	700	400	

(i) S_{ij} = Unknown quantity

$19S_{11}D_1 + 30S_{12}D_2 + 50S_{13}D_3 + 10S_{14}D_4$

$70S_{21}D_1 + 30S_{22}D_2 + 50S_{23}D_3 + 10S_{24}D_4$

$40S_{31}D_1 + 8S_{32}D_2 + 70S_{33}D_3 + 20S_{34}D_4$

(ii) The objective is to minimize (Z)

(iii) Create a constraint.

$S_{11}D_1 + S_{12}D_2 + S_{13}D_3 + S_{14}D_4 = 700$

$S_{21}D_1 + S_{22}D_2 + S_{23}D_3 + S_{24}D_4 = 900$

$S_{31}D_1 + S_{32}D_2 + S_{33}D_3 + S_{34}D_4 = 500$ } Supply Constraint

$S_{11}D_1 + S_{21}D_1 + S_{31}D_1 = 500$

$S_{12}D_2 + S_{22}D_2 + S_{32}D_2 = 800$

$S_{13}D_3 + S_{23}D_3 + S_{33}D_3 = 700$

$S_{14}D_4 + S_{24}D_4 + S_{34}D_4 = 400$ } Demand Constraint

$S_{ij} \geq 0$ for i, j (non negativity function)

i = row & j = Column

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

(i) NWC

	D_1	D_2	D_3	D_4	SS
S_1	19	30	50	10	0
S_2	70	30	40	60	0
S_3	40	8	70	20	0
DD	500	500	700	1400	3400

$TC = (500 \times 19) + (200 \times 30) + (600 \times 30) + (300 \times 40) + (400 \times 70) + (1400 \times 20) = 101,500$

(ii) Least Cost Method (LCM)

	D_1	D_2	D_3	D_4	SS
S_1	19	30	50	10	700
S_2	70	30	40	60	200
S_3	40	8	70	20	300
	200	500	700	1400	3400

$TC = (700 \times 10) + (200 \times 70) + (700 \times 40) + (300 \times 40) + (500 \times 8) + (700 \times 20) =$

(iii) Vogel Method.

	D_1	D_2	D_3	D_4	SS
S_1	19	30	50	10	700
S_2	70	30	40	60	200
S_3	40	8	70	20	300
	500	500	700	1400	3400

$TC = (500 \times 19) + (200 \times 10) + (700 \times 40) + (200 \times 60) + (500 \times 8) + (1000 \times 20) =$

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

16/07/018. Markov Decision Model.

Markov decision Model is a system that changes per time.

Introduction:-

Many Management problems can be modelled by using Markov Chain. A Markov chain is a probabilistic device that is useful for representing system that changes overtime.

In our discussion, we shall illustrate their use in analysing sound problem, by using an example.

Learning Outcomes.

At the end, you should be able to:

1 Give the definition of Markov chain
2 List & explain the terms use in Markov chain
3 Draw & explain the transition diagram
4) Derive the transition matrices from the transition diagram
5) Explain what's meant by step transition.
6 Compute the steady state probability
7 - future Market Share
8 Calculate the (average) Mean return time

The Branch Switching Model.

Mr. Jones, the owner of a large Supermart has been considering the branch switching behaviour of his Customer. In a purchase

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

of 2 products say product A & B, each are alternative substitute to one another. The marketing staff of the supermarket as collected data over a period of time that gives the following trend in the pattern of the branch switching of the customer.

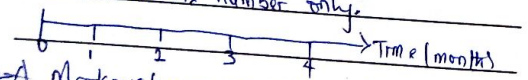
(i) Customer that purchase product A at the beginning of 1mth will also purchase product A at the beginning of the following month with a probability of 0.6 while he will switch to product B with the probability of 0.4

(ii) A customer that purchase product B at the beginning of 1mth will then purchase product B at the beginning of the following month with a probability 0.7 while the customer will switch over to product A at the beginning of another month with the probability of 0.3

Mr. Jones will like to use the above information to generated & analyse buyer behaviour patterns in the future.

Terms of Markov Chain.

(1) Time:- In Markov chain, we refer to the specific time & time can be represent as an horizontal time axis. Action is also place at distinct whole number only.



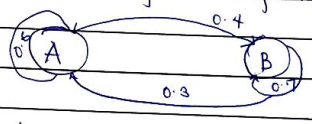
A Markov chain is nothing but a subset of a markov process.

(2) State:- This simply indicates the particular product a customer is purchasing at a particular point in time.

(3) Transition:- This simply means movement of changing state.

(4) State Transition:- This is the movement from one state to another state which may also be the state one was.

(5) Transition Diagram:- In regard to the illustration.



In a case where 3 products are involved, we will have 3 circle in our TD.

(6) Transition Matrix

Matrix is the arrangement of number in roles & Column.

The transition matrix of the illustration is

This Month →	A	B
A	0.6	0.4
B	0.3	0.7

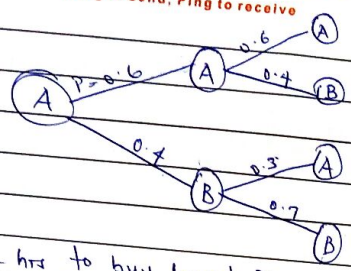
= P'

A = Last Month, B = This Month

Example:- If a customer initially buy brand A, what is the probability that two month wants to buy brand A. What is his probability of buying A.

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace
...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive



$(0.6 \times 0.6) + (0.4 \times 0.3)$

$0.36 + 0.12$

$= 0.48.$

∴ if his to buy brand B in 2 months.

$(0.6 \times 0.4) + (0.4 \times 0.7)$

$0.24 + 0.28$

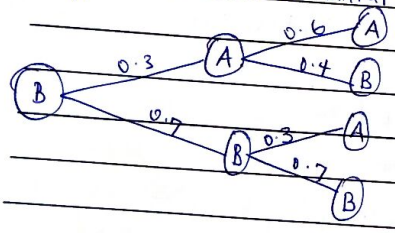
$= 0.52.$

2 Months.

This Month	A	B
A	0.48	0.52
B	0.39	0.61

$= P^2$

Suppose the customer initial purchase is B.



$(0.3 \times 0.6) + (0.7 \times 0.3)$

$0.18 + 0.21$

$= 0.39$

$(0.3 \times 0.4) + (0.7 \times 0.7)$

$0.12 + 0.49$

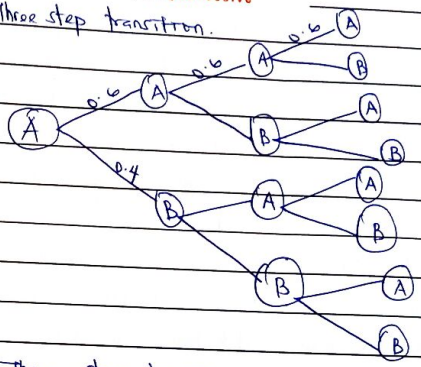
$= 0.61$

The above is referred to as 2-step transition.

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

Three step transition.



Three step transition is solved by bi-matrix multiplication. $\Rightarrow P^3 = P^2 \cdot P$

$$P^2 = P \cdot P = \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

	A	B
A	$0.6\times0.6 + 0.3\times0.4$ $0.36 + 0.12$	$0.4\times0.6 + 0.7\times0.4$ $0.24 + 0.28$
B	$0.6\times0.3 + 0.3\times0.4$ $0.18 + 0.21$	$0.4\times0.3 + 0.7\times0.7$ $0.12 + 0.49$

$$P^2 = \begin{bmatrix} 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix}$$

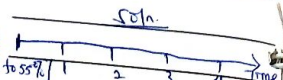
$$P^3 = P^2 \cdot P = \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix} \begin{bmatrix} 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix}$$

But in Markov, $P^3 = P \cdot P^2$ or $P^2 \cdot P$ note, only one is right & the

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

Soln.



right expression of $P^3 = P^2 \cdot P$ Initial take up time is represented as $[S_A \; S_B]_0$ that will be the market share in the 3 months.

$$P^3 = \begin{bmatrix} 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix} \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.288 + 0.156 & 0.312 + 0.244 \\ 0.144 + 0.273 & 0.156 + 0.427 \end{bmatrix}$$

$$P^3 = \begin{bmatrix} 0.4448 & 0.556 \\ 0.417 & 0.583 \end{bmatrix}$$

Markov Chain.

Definition:- A Markov chain is a process in which the probability of the occurance of an events in the future depends only on current events and not that regularity on previous events.

Market Share Analysis.

Suppose that the share of the Market controlled by product A & B initially is 45% to 55%, then you can use this information two product for review to forecast the share of the market that will be controlled in the future.

$$[S_A \; S_B]_2 = [S_A \; S_B]_0 P^2$$

$$\begin{bmatrix} 0.45 & 0.55 \end{bmatrix} \begin{bmatrix} 0.48 & 0.52 \\ 0.39 & 0.61 \end{bmatrix} = \begin{bmatrix} 0.216 + 0.2145 & 0.2340 + 0.3355 \end{bmatrix}$$

$$\begin{bmatrix} 0.4305 & 0.5695 \end{bmatrix}$$

$$[S_A \; S_B]_2 = [0.4305 \;\; 0.5695]$$

Inventory Analysis.

Suppose we have 5,000 customers in the purchase of the market, two product every month with each customer buying a packet of either product A or B. What should be the inventory of the two product for the month.

Soln.

Inventory for $A = 0.4305 \times 5000$

$= 2152.5$

Inventory for $B = 0.5695 \times 5000$

$= 2847.5$

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

Steady state.

Limit $n \to \infty \; P^n = [\pi_A \; \pi_B]$ with

$\pi_A + \pi_B = 1$

$[\pi_A \; \pi_B] = [\pi_A \; \pi_B] P$

where P is the initial transition matrix

$$= [\pi_A \; \pi_B] \begin{bmatrix} 0.6 & 0.4 \\ 0.3 & 0.7 \end{bmatrix}$$

$$[\pi_A \; \pi_B] = [0.6\pi_A + 0.3\pi_B \quad 0.4\pi_A + 0.7\pi_B]$$

$\pi_A = 0.6\pi_A + 0.3\pi_B$ ——eqn(i)

$\pi_B = 0.4\pi_A + 0.7\pi_B$ ——eqn(ii)

$\pi_A + \pi_B = 1 \Rightarrow \pi_A = 1 - \pi_B$

$1 - \pi_B = 0.6(1 - \pi_B) + 0.3\pi_B$

$1 - \pi_B = 0.6 - 0.6\pi_B + 0.3\pi_B$

$1 - 0.6 = \pi_B + 0.3\pi_B - 0.6\pi_B$

$0.4 = 1.3\pi_B - 0.6\pi_B$

$0.4 = 0.7\pi_B$

$\pi_B = 0.4$

$0.7 = 4/7.$

Since $\pi_A = 1 - \pi_B$

$\pi_A = 1 - \frac{4}{7}$

$\pi_A = 3/7.$

	A	B
A	3/7	4/7
B	3/7	4/7

Mean Return Time.

Mean return time for A = 1 Note: 1 = 1 month

$\pi_A = \frac{1}{3/7} = \frac{7}{3}$ months.

$B = \frac{1}{\pi_B} = \frac{1}{4/7} = 7/4$ months.

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

23/01/18:

There are 3 decision making environment which are:

1) Decision making under Certainty
   - Stochastic/under this we have risk.
   - Uncertainty

In every decision making, there are decision variables. Decision variables are factors that influence decision making & there's 2 type
1) Controllable decision variable.
2) Uncontrollable

Controllable decision variable are variable that can be controlled by the decision maker.

Replacement Analysis.

Organizations exist in some technical context that are made up of human and non-human items. However, because nothing last forever, there is a need to replace or maintain the item at some point in time.

Replacement simply mean substituting for or [illegible] back.

While replacement analysis is the process by which various Cost Consequences involved in replacement are studied to identify or make optimal replacement decision.

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

In order to do this, there is a need to know the categorized of items that exist in an organization.

Categories of Items in an Organization.

Basically, there are 3 categories
(1) Sudden failure items:- They are items that works fine up to a certain time and fail e.g bulb, fan belt.
(2) Items that deteriorate/Items that deprecate with time or usage but the life can be dampled if they are maintained e.g Air conditional
(3) Items that deteriorate/deprecate in efficiency until it fails eg human.

Assumption of Replacement Analysis.

(1) The objective is to minimize Cost.
(2) Constant replacement Cost.
(3) Constant technological level
(4) - Maintenance cost
(5) - output quality
(6) - Performance level.

Cost Associated with Replacement Analysis.

(1) Sudden failure items:
(i) replacement Cost:- This is the purchase price of the item.
(ii) Consequential Cost of failure:- Cost incurred when item fail suddenly.
(iii) labour cost
(iv) Cost of replacement:- This is the summation of all cost involved in replacing the item.

(2) Deteriorative Items:
(i) Maintenance Cost
(ii) depreciation

Replacement [illegible]

(1) Sudden failure Items:

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

There are 2 types of policies for Suddenfarture items viz:
i Individual replacement policy
ii group replacement policy

Individual replacement is when you change the item as they go bad one-by-one but group replacement is when you change the all whole item at once and group replacement policy is divided into 2 viz: periodic group replacement

Mechanism for Individual Replacement Policy.

Step 1: find the prob. of failure per Period $\Rightarrow P_i$

Note:- When prob total is greater than 1 it means its in its cummulative form, then you need to decummulate before solving.

Step 2: Find the average life of item $\Rightarrow A_L = \sum P_i t_i$

Step 3: Find the number of items to be replaced $N_r = \frac{N_0}{A_L}$

where N_0 = initial num. of item
A_L = average life

Step 4:- Find the cost of individual replacement $\Rightarrow I_c = N_r(ic)$

where ic = Purchase price + Consequential cost of item + labour i.e $ic = P + Cosc + L$.

P = replacement cost / purchase cost

Mechanics of Group Replacement Policy.

Step 1: Tabulate the time period

Step 2: Find the prob. of failure per period i.e P_i

Step 3: Find the number to be replaced per period using:-

$N_1 = N_0 P_1$

where $N_0 \rightarrow$ Initial [illegible] of item

$N_2 = N_0 P_2 + N_1 P_1$

$N_3 = N_0 P_3 + N_1 P_2 + N_2 P_1$

$N_4 = N_0 P_4 + N_1 P_3 + N_2 P_2 + N_3 P_1$

$N_5 = N_0 P_5 + N_1 P_4 + N_2 P_3 + N_3 P_2 + N_4 P_1$

Step 4:- find the cummulative no to be replaced per period.

Step 5: Find the cummulative individual related cost per period: This is equal to cummulative no to be replaced per period x individual replacement cost per unit

Step 6:- Find fixed cost i.e Initial number of items x group replacement cost.

Step 7:- Find the total cummulative replacement cost = Cummulative individual replacement cost per period + fixed cost per period.

Step 8: find the average total cummulative replacement cost per period = Total cummulative replacement cost per period ÷ the respective period.

Step 9: Select the least average total cummulative replacement cost & conclude that group replacement should be done at the end of that period.

Step 10: Note:- Only do step 10 when ask to advice or Compare the policy. Compare the cost of individual replacement policy with the cost of group replacement policy & recommend the least cost policy option.

Read maintenance policy for deteriorating item

(1) Minimal average cost method: apply the algorithm.
(2) Differential cost method: go through FBA 220 & Calculus.

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

Example: Assuming you're the maintenance officer for the faculty of MS & you've asked to recommend
i. the appropriate RP to be adopted
ii. to replace the florisent bright bulb in the faculty classroom block comprising of Rm 1, 2, 3, 4, 5, 6, 7, 8, 9.

Assuming further that the initial num. of florisent bright bulb in all the classroom is 200 & the cost of replacing them individually as they fail is ₦3,500 while the cost of replacing them as a group is ₦10,000. You're required to analyse the situation completely given the following failure data per month.

Month	Prob. of failure
1	0.03
2	0.30
3	0.55
4	0.90
5	1.00

Soln.

We have to decummulate because the total of the prob. is more than 1 i.e ∑ sum upto 2.75.

Month	Prob. of failure P_i	$P_i t_i$
1	0.03	0.03
2	0.30	0.27
3	0.55	0.25
4	0.90	0.35
5	1.00	0.10
	2.75	1.00 ≈ 3.22

Computing Individual Replacement Policy

(1) Find P_i per period

(2) $A_L = \sum P_i t_i = 3.22$

(3) $N_r = \frac{N_0}{A_L}$

Where $N_0 = 200$

$N_r = \frac{200}{3.22}$

$N_r = 62.11 \approx 62$

(4) $T_C = N_r (i_c)$

$= 62 \times 3500$

$= ₦217,000$

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

30/01/18

When dealing with deteriorating items, there are different models that can be used to determine the optimal point in time for replacement and 2 of these methods are:-

1) Minimum average cost method
2) Calculus method

Example:- The faculty of Magt. Sci. has inverter sets in its classroom blocks. The purchase price of the inverter set is ₦10M, the re-sale value & Maintenance cost over a 5yr period is given in the table below

Yr	Resale value ₦'000	Maintenance cost ₦'000
1	8,000	300
2	6,700	500
3	5,000	800
4	2,400	1,400
5	1,200	3,000

Required:- Find the optimum time

that the faculty of Magt. Sci. should replace the inverter set without taking into cognisance the cost of Capital. Use the average total cost method to solve the problem and the alternative solution can be achieved by using the Differential Calculus method. This entails differentiating the average total cummulative Maintenance cost (y) with respect to the life span of the item (x) and then apply the turning point rule to solve the differentiated equation with respect to x.

Example:-

Assuming the annual average total cummulative cost (y) of the inverter set in the faculty of Magt. Sci. expressed as a function of their life span is given by the equation

$y = 5x^2 - 60x$

Required:- Determine the optimal

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

replacement age for the inverter set using the Calculus Method.

Soln.

$y = 5x^2 - 60x + 2,000$

$\frac{dy}{dx} = 10x - 60 + 0$

at turning point,

$\frac{dy}{dx} = 0$

$\therefore 0 = 10x - 60$

$\frac{60}{10} = \frac{10x}{10}$

$x = 6yrs.$

$\therefore$ the inverter sets is to be replace after 6yrs.

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

Assembly Line Balancing.

Organizations use various production system to ensure that goods & services that meet customers needs are produced in order to do this, they can apply either the "flow process system" or the "job process system". The "flow process" is a production system in which materials "flow" uninterrupted from the point of entry into system to the point of exit while a "job system" is a production system in which goods are produced based on customer's specification irrespective of the production process adopted. Sometimes it balances in capacity and resource utilization are experienced. Hence, there is a need to embark on assembly line balancing.

Assembly line balancing seeks to distribute task to work stations so as to optimise line operations. An assembly line is an arrangement of organizational resources whereby a product is assemble through the performance of the specific sucessive operation on incomplete unit as its passes by in a series of stages organized in a direct line. This stages are called work stations.

Assembly lines are mostly used to assemble complex products like bicycles, electronics, household appliances, etc. Though, most assembly line method are developed mainly for manufacturing processes but they can also be applied in other areas of business that deals with project Mgt.. The decision problem of assembly line balancing entails assigning an appropriate work allocation to the work stations so as to minimise idleness and maximise efficiency.

The goal of assembly line balancing is to find ways of equalizing performance time at all work stations.

Note:- It is not possible or rarely possible to have a perfectly balanced line.

Advantages.

1. It reduces variation.
2. It minimises idle time.
3. It minimises waste.
4. It increases efficiency.
5. It reduces over burden.

Line Balancing Solution Techniques.

There are several line balancing techniques but they can be categorised into two viz:

1. Mathematical method:- They are exact or precise approaches which gives optimal satisfaction.
2. Heuristic Method:- They are method based on the use of thumb.

The most popular one is the 'Waters and Trill bridge method'

Operationalization of Terms

(1) Operations:- This are tasks to be done. They are the smallest unit into which the total 'job content' can be divided. They are denoted by "i"

(2) Cycle time:- This is denoted by "C" and $C = \frac{T}{n_o}$. It is the maximum time available for a task or group of tasks to be completed at each work station. Where:-

T = time to complete all jobs in a day.
n_o = number of items to be produced daily
t_i = time taken to complete task "i"
S_j = total time taken to complete work stn
$S_j = C \Rightarrow$ perfectly balanced line.

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

$\sum t_i < C \Rightarrow$ actual balance time

situation.

$S_E = \frac{\sum t_i}{N_a \times C} \times 100$

S_E = System efficiency

= System efficiency Measure.

This tells us the level of the efficiency of the system. It is obtain using the above formula.

Mechanics of Line Balancing:

Since in real time, we have fairly balance situation, the Wester & Hilbridge realistic method will be adopted. Below is the modified algorithm.

Step 1: define all the elemental task involved in getting a job done.

Step 2: Step the precedent relationship

Step 3: draw the network diagram

Step 4: Calculate the allowable or required work station Cycle time using the formula $C = \frac{T}{No}$.

Step 5: Compute the theoritical minimal number of work station needed using the formular:

$N = \frac{\sum t_i}{C}$

Step 6: Construct the number of following task table.

Note:- In drawing the work network we can use:

(i) activity on the arch

(ii) ⌐ on note and this is use when applying the Wester & Hilbridge method.

Step 7: Construct assembly line balancing allocation or assignment table using the following headings:

Note:- Balance delay refers to idleness and it is a measure of system inefficiency as denoted as BD.

Step 8: Apply assignment rule as follows:

(i) Primary rule:- Assign the task with the ... of follow...

(ii) Secondary rule:- In case of a tie, assign the task with the longest operation time.

Step 9: repeat step 8 until tasks have been assigned.

Step 10: evaluate system efficiency using the following formula

$S_E = \frac{\sum t_i}{N_a \times C} \times 100$.

Step 11: re-balance if necessary by going back to step 1.

Example:- Malcom BHD has a assembly plant where rechargable lambs are manufactured on a daily basis. Assume that her assembly plant operates a 7hr working day.

Required:

(a) find the balance that minimizes the number of work stations assuming she produces 630 lamps.

work stations	Task	Task time t_i	No of following task	Task with most following task	Task with longest operation time.

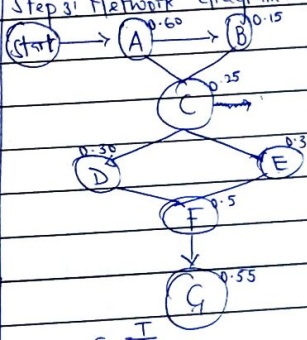
...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

(b) Compute balance delay given the following table:

Tasks	Description	Immediate predecessor relationship	Time
A	Get Component	—	0.6
B		A	0.15
C	insert fan blade	A, B	0.25
D	insert red light	C	0.30
E	Connect wires	C	0.35
f	Couple all parts	D, E	0.50
g	label	F	0.55
			2.70

Soln.

Step 1: Network diagram.



Step 1: $C = \frac{T}{No}$

$T = 7hrs \Rightarrow 7hrs \times 60mins$

$T = 420mins$.

$C = \frac{420mins}{630} = 0.67mins$

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

Step 5: $N_A = \frac{\Sigma t_i}{C}$

$\Sigma t_i = 2.7$

$N = \frac{2.7}{0.67\text{min}} = 4.03$

$N \approx 5$ work station.

Step 6:

Tasks	Time	No of followers
A	0.60	6
B	0.15	5
C	0.25	4
D	0.30	2
E	0.35	2
F	0.50	1
G	0.55	0

Step 7:

Work station	Task	Task time t_i	No of following task	Idle time	Feasible remaining task	Task with most num. of follower	Task with longest operation time
1	A	0.60	6	0.07min	—		
2	B	0.15	5	0.52	C, D, E, F	C	
	C	0.25	4	0.27min			
3	E	0.35	2	0.32	D		D, E
	D	0.30	2	0.02min			
4	F	0.50	1	0.17min			
5	G						

idle time = $C - t_i$

Step 10:

$E = \frac{\Sigma t_i}{N_A \times C} \times 100$

$= \frac{2.70}{5 \times 0.67} \times 100$

$= 80.60\%$

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

06/02/18: Sequencing & Routing

Sequencing:-

Introduction:- Every organisation will like to utilize its productive system effectively & efficiently so as to maximise profit by meeting delivery deadlines. Thus, in many operations have to be performed and where as, there are limited resources in terms of plant & machines on which these job have be performed.

Definition:

Sequencing is the selecting of appropriate order for a series of jobs to be done on a finite number of service facilities.

Assumptions:

(1) All machines can process only one job at a time.

(2) Processing time is known & fixed.

(3) When a job starts, it must be completed.

(4) Preceeding operations must be completed befor the succeeding one.

(5) All jobs are in the specified order as soon as possible.

(6) All jobs are ready for processing.

(7) All machines have different capacity and capability.

(8) Processing time of a job have no relationship with the order in which job is processed.

(9) No time is wasted in shifting a job from one machine to another i.e wasted time is zero.

Type of Sequencing Problem

(1) Type I:- n jobs on one machine.

(2) Type II:- n jobs on Two machines.

(3) Type III:- n jobs on M machines where $M > 1$

Sequencing [illegible] jobs.

Machines:- This facility which has some processing capability.

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

b) Scheduling:- This is when time & Sequencing of processing a job is Specified.

c) Total elapsed Time:- It is the time that lapsed between the starting of the first job and the finishing of the last job.

d) Idle time:- This is the time the facilities or machines are not used during the total elapsed time.

e) No passing rule:- This refers to the rule of maintaining the order in which jobs are to be processed on a given machine i.e a particular job must first be complemented before any other one can start.

Examples: The processing time for Jobs A, B, C, D, E and the due dates for these job in a machine shop are given in the table below!

Jobs	Processing Time	Due date (Day)
A	9	16
B	7	20
C	5	25
D	8	20
E	6	40

The job are to be sequenced according to!

(1) First Come first Serve (FCFS)
(2) Shortest processing time (SPT)
(3) Least slack (LS)
(4) Expected due date (EDD)

Soln.

FCFS.

Step 1: Repeat the first two Columns.

Jobs	Processing Time
A	9
B	7
C	5
D	8
E	6

Step 2: Cummulate the ... time and add them together as it is called the flow time.

Jobs	Process time	flow time
A	9	9
B	7	16
C	5	21
D	8	29
E	6	35
		110.

Step 3: Repeat due date.

jobs	Process time	flow time	due date
A	9	9	16
B	7	16	20
C	5	21	25
D	8	29	20
E	6	35	40
		110	

Step 4: Do some comparing between the flow time with the due date by checking for the difference.

Job A = No lateness
Job B = ✓
Job C = ✓
Job D = lateness
Job E = No lateness

Jobs	Process time	flow time	due date	lateness
A	9	9	16	0
B	7	16	20	0
C	5	21	25	0
D	8	29	20	9
E	6	35	40	0
		110		9

Using FCFS

(1) Total Completion time will be 35days.

(2) Average Completion time is equal to total flow time ÷ no of items.
i.e $\frac{110}{5} = 22$ days.

(3) Average number of work at the Centre = Total flow time ÷ total process time.
$\therefore \frac{110}{35} = 3.1$ jobs.

(4) Average job late = total lateness ÷ num. of jobs. = 1.8 days

(5) Number of late job = 1, job D.

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

13/02/18 Least slack.

Step 1: Difference between due date & processing time gives the slack

Example:-

Job	Due date	Processing time	slack
A	16	9	7
B	20	7	13
C	25	5	20
D	20	8	12
E	40	6	34

Step 2:- Arrange the job, processing time and due date on the ascending order of the slack.

Job	Processing time	due date	slack
A	9	16	7
D	~~8~~	20	12
B	~~7~~	20	13
C	~~5~~	25	20
E	~~6~~	40	34

Step 3: Delete the slack and cummulate the processing time and put the values in between processing time and due date and tag it flow time & add up the flow time.

Job	Processing time	flow time	due date.
A	9	9	16
D	8	17	20
B	7	24	20 (4)
C	5	29	25 (4)
E	6	35	40
	35	144	

Step 4: Determine the late jobs by checking the due date with flow time. Using least Squares (5)

(1) Total Completion time = 35 days

(2) Average = Total flow time / no of jobs

= 144 / 5 = 22.8 days

(3) Average no of work of the centre

= Total flow time / total process time = 144 / 35 = 3.26 days

(4) Average job late

= Total lateness / no of job = 8 / 5 = 1.6 days

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

(5) late Jobs (Tardy) = 2 jobs B & C

Expected due Date.

Step 1: Arrange due date from least to the highest with their corresponding jobs in that order.

Job	Due date
A	16
B	20
D	20
C	25
E	40

Step 2: put their corresponding processing time

Jobs	Processing time	due date
A	9	16
3	7	20
D	8	20
C	5	25
E	6	40
	35	~~85~~ 121

Step 3: Cummulate process times & put the values between the process time & due date and add up the flow time as follows:

Job	Processing time	flow time	due date
A	9	9	16
B	7	16	20
D	8	24	20
C	5	29	25
E	6	35	40
	35	113	

Step 4: Determine the late jobs by checking the due date with the flow time.

Job	process time	flow time	Due date	lateness
A	9	9	16	0
B	7	16	20	0
D	8	24	20	4
C	5	29	25	4
E	6	35	40	0

Using FIFO,

(1) Total Completion Time = 35 days

(2) Average Completion = 113/5 = 22 days

(3) Average job late = 8/5 = 1.5 days

(4) Average no of work of the Centre

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

$= \frac{113}{35} = 3.23$ jobs.

(5) Thirty jobs - 2 days.

16/02/018 Project Management.

Overflow of Project Management

Project Manage is preferred to General operation Management because:

1) Limited time frame
2) Specific objective
3) Bureaucracy

Key factors or Matrix of PM

(1) Time
2) Cost and
3) Performance objectives

PM over General operational Management

Project Mgt. is use because:

1) There are special needs that don't lend themselves to functional Management.
2) There are pressure for new product.
3) PM is use for cost reduction.

Key Critical Sucess factors.

(1) Top-down Commitment
(2) Respected & Capable project Manager.
(3) Careful tracking & Control.
(4) Good Communication.
(5) Enough time to plan.

Main Administrative Issues.

(1) Executive representation
(2) Project selection

Tools of Project Management.

(1) Work break-down structure
(2) Network diagram
(3) Gantts chart
(4) Risk analysis
(5) Risk Management

Key Decisions of PM

(1) Decide which project to implement.
(2) Select the project Manager
(3) ✓ ✓
(4) Plan & design the
(5) Manage & Control the project ie Monitoring
(6) Decide if & when the project is to be terminated.

Project life cycle.

There are 4 phases:

(1) definition of the project
(2) planning of the project by spelling out all the details of the work to be done for the project
(3) Execution i.e the work is done
(4) (5) Termination or delivery.

Note:- These 4 phases overlap.

Example: A bank wants to open a new branch.

Activity	0	2	4	6	8	10	12	14	16	18	20
Locate new facilities	▨	▨	▨	▨	▨						
Interview prospect staff			▨	▨							
Hire & train					▨	▨	▨	▨	▨		
Select another furniture							▨	▨	▨		
Remodel of the building							▨	▨	▨	▨	
Furniture refuel and set up									▨	▨	
Move in & start up										▨	▨

Activity:
1 Locate new facilities Wk 0-8
2 Interview prospect staff Wk 0-4
3 Hire & train Wk 4-13
4 Select another furniture Wk 8-14
5 Remodel of the building Wk 8-18
6 Furniture refuel & set up Wk 12-16
7 Move in & start up Wk 18-20

Network diagram is a diagram of project activities that shows sequential relationship i.e how they follow each other.

A path is a sequence of activity that leads from the starting node to the finishing node.

Gantt Chart

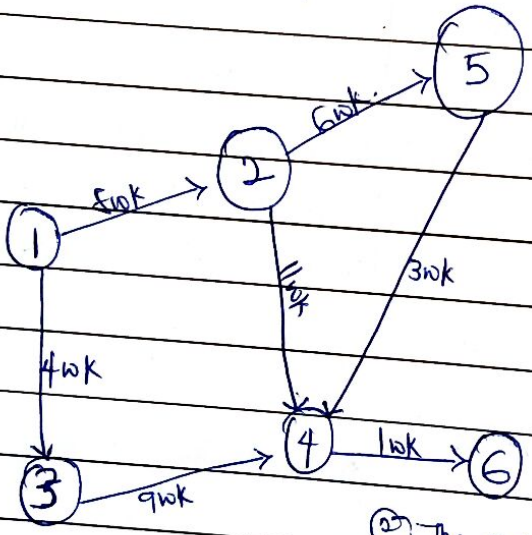
...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace

pingexpress
Ping to send, Ping to receive

Determine!
(1) The length of each part.
(2) The Critical path.
(3) What is the expected duration of the project?
(4) What is the amount of slack?

The Critical path is the longest path that determines project duration.

Slack is the difference between the Critical path and the path.



(1) Path A = 1-2-5-4-6
✓ B = 1-2-4-6
✓ C = 1-3-4-6

length of each paths.
Path A = 8+6+3+1 = 18wks
✓ B = 8+11+1 = 20wks
✓ C = 4+9+1 = 14wks.

(2) The Critical path is path B.

(3) The expected duration of the project is determined by the Critical path and it is = 20

(4) Slack!
Path A = 20-18 = 2wk
✓ B = 20-20 = 0wk
✓ C = 20-14 = 6wks.

...Ping Express - instant transfer to bank accounts while putting a smile on a face.
#PutASmileOnAFace